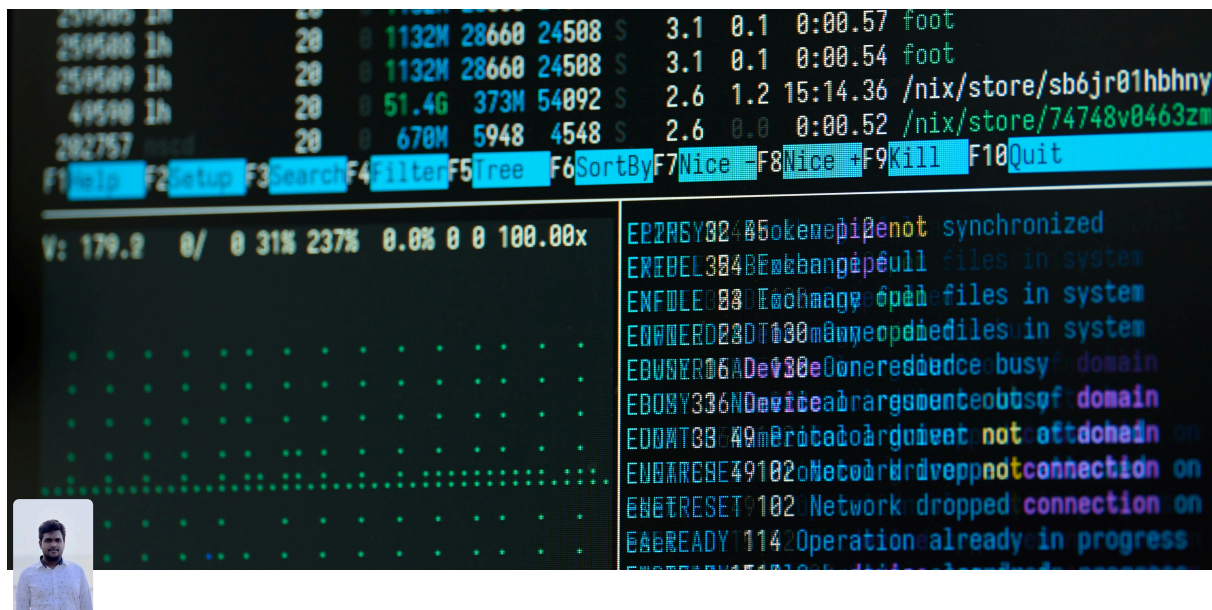




Real-Time Intrusion Detection & Log Analysis

1. Project Overview

- **Project Name:** Real-time SSH Brute Force Detection System
- **Category:** Defensive Security / Threat Hunting
- **Author:** Isuru Udana Pintu
- **Target Environment:** Kali Linux

2. Objective

- To develop a lightweight, proactive monitoring tool that detects unauthorized access attempts in real-time.
- This project simulates the core activities of a **Security Operations Center (SOC)** or a **Threat Hunting** team.

3. Technical Stack

- **Log Aggregation:** `journalctl` (Systemd log manager).
- **Pattern Matching:** `grep` and `awk` for string filtering.

- **Protocol:** SSH (Secure Shell).

4. Implementation (The Script)

```
#!/bin/bash

#####
#####
# Script Name:    log_monitor.sh
# Description:    Real-time SSH Brute Force Detector.
# Author:         Isuru Udana Pinthu
# Date:           2026-02-26
# Version:        1.0
# Usage:          ./log_monitor.sh
#####
#####

echo "=====
=====
echo "      REAL-TIME SSH INTRUSION DETECTION SYSTEM
"
echo "=====
=====
echo "[+] Monitoring system logs for Brute Force attack
s..."

# Using journalctl to follow SSH logs in real-time
sudo journalctl -u ssh -f | while read LINE; do
    # Looking for 'Failed password' in the live log stream
    if echo "$LINE" | grep -q "Failed password"; then
        echo -e "\e[31m[ALERT]\e[0m Failed login attempt de
tected!"
        echo "Log Detail: $LINE"
    fi
done
```

5. Testing & Validation

- **Attack Simulation:** Triggered multiple failed SSH login attempts via `localhost`.
- **Detection Logic:** The script correctly identified the "Failed password" string from the system journal.
- **Strategic Value:** Demonstrates the ability to parse complex system logs to extract **Indicators of Compromise (IoCs)** and provide immediate visual alerts to administrators.

6. Evidence

```

kali@kali: ~/Desktop/Projects/CTI
Session Actions Edit View Help

(kali@kali)~[~/Desktop/Projects/CTI]
$ nano asm_scanner.sh

(kali@kali)~[~/Desktop/Projects/CTI]
$ nano log_monitor.sh

(kali@kali)~[~/Desktop/Projects/CTI]
$

Shell No. 1
Session Actions Edit View Help

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# Usage:          ./log_monitor.sh
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echo "===== "
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echo "[+] Monitoring system logs for Brute Force attacks..."

# Using journalctl to follow SSH logs in real-time
sudo journalctl -u ssh -f | while read LINE; do
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    if echo "$LINE" | grep -q "Failed password"; then
        echo -e "\e[31m[ALERT]\e[0m Failed login attempt detected!"
        echo "Log Detail: $LINE"
    fi
done
~
~
~

```

